

# DIGITAL MEASUREMENT AND CONTROL



# LECTURE 1: INTRODUCTION

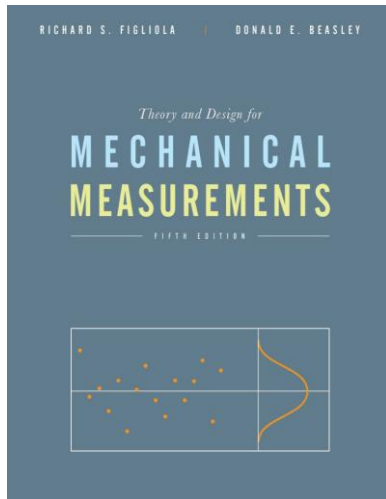
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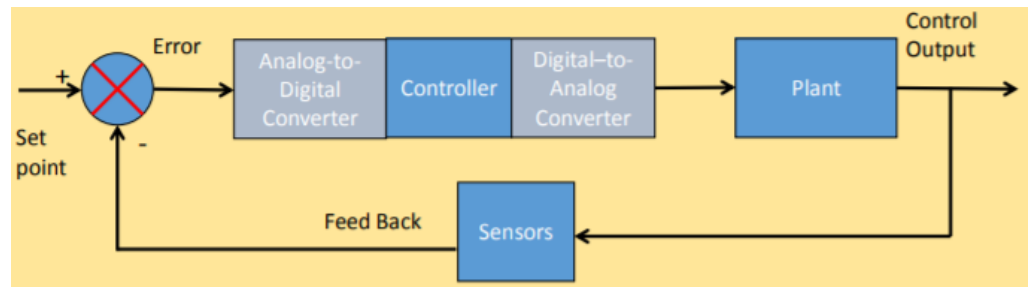
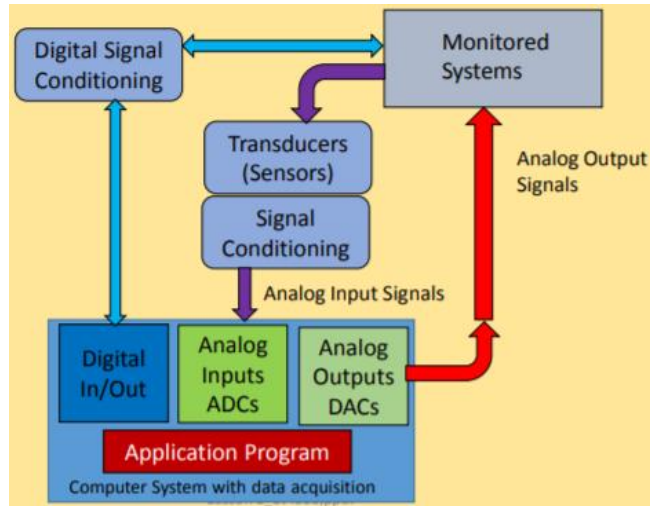
# Course Overview

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- > Lecturer: Hoang Gia Hung
  - R702, E3 building (by appointment).
- > Textbooks:
  - Figliola and Beasley "*Theory and Design for Mechanical Measurements*" 5th Ed., John Wiley & Sons, 2011.
- > Labs:
  - NI Elvis II+ DAQ & LabView
- > Grading:
  - Labs - 20%
  - Midterm - 20%
  - Final project - 60%

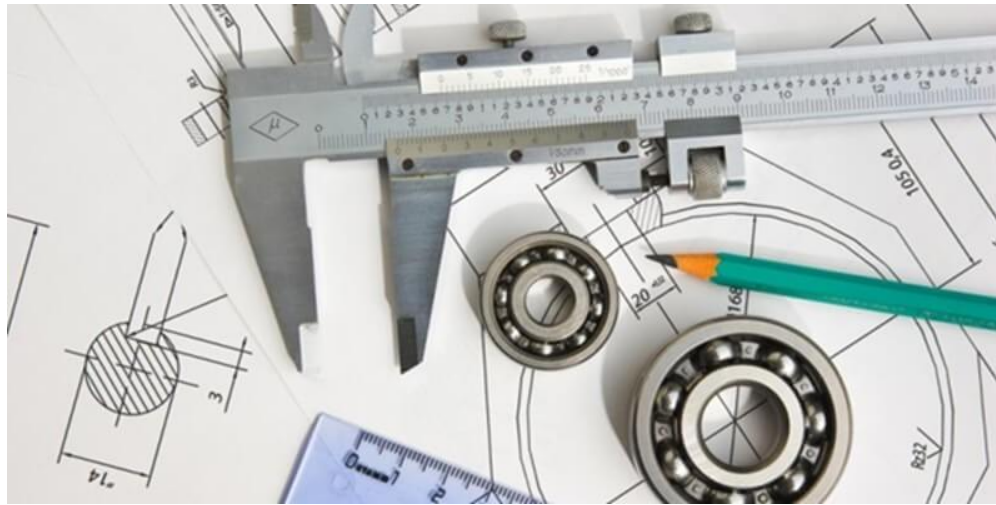
# Course Objectives



1. Understand fundamental principles of digital measurement
  - Able to design engineering tests & analyze test data
2. Design and implement a real-world digital control system
  - Step-by-step lab works
  - Final project

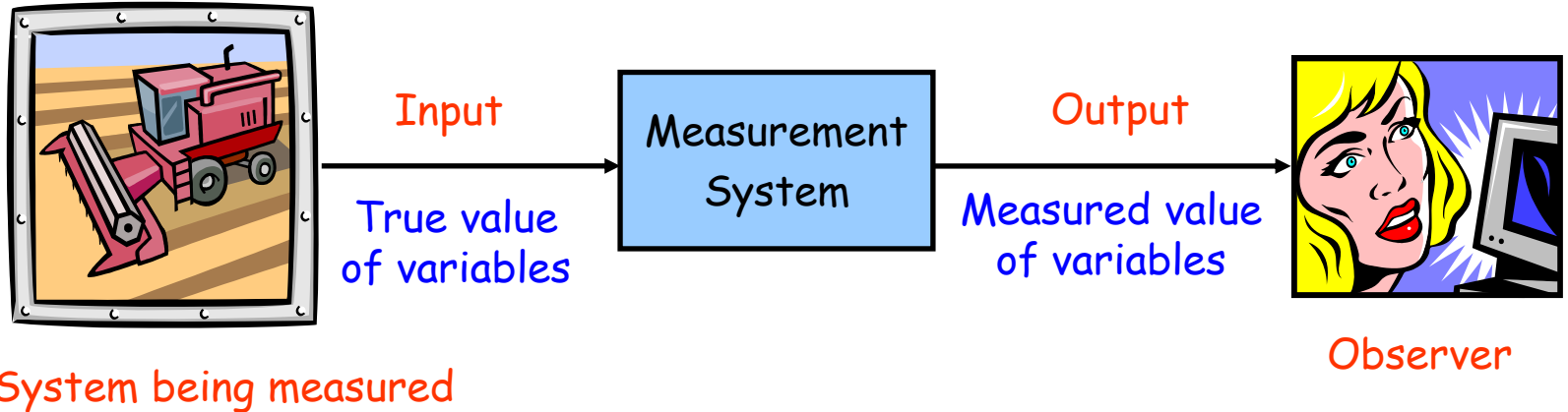
# Why Do We Measure?

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- > Engineering decisions depend on numbers
  - Is the system stable?
  - Is temperature safe?
  - Is voltage within tolerance?
  - Is performance acceptable?
- > No measurement → no engineering.

# What Is Measurement?



## > We never observe the “true value”

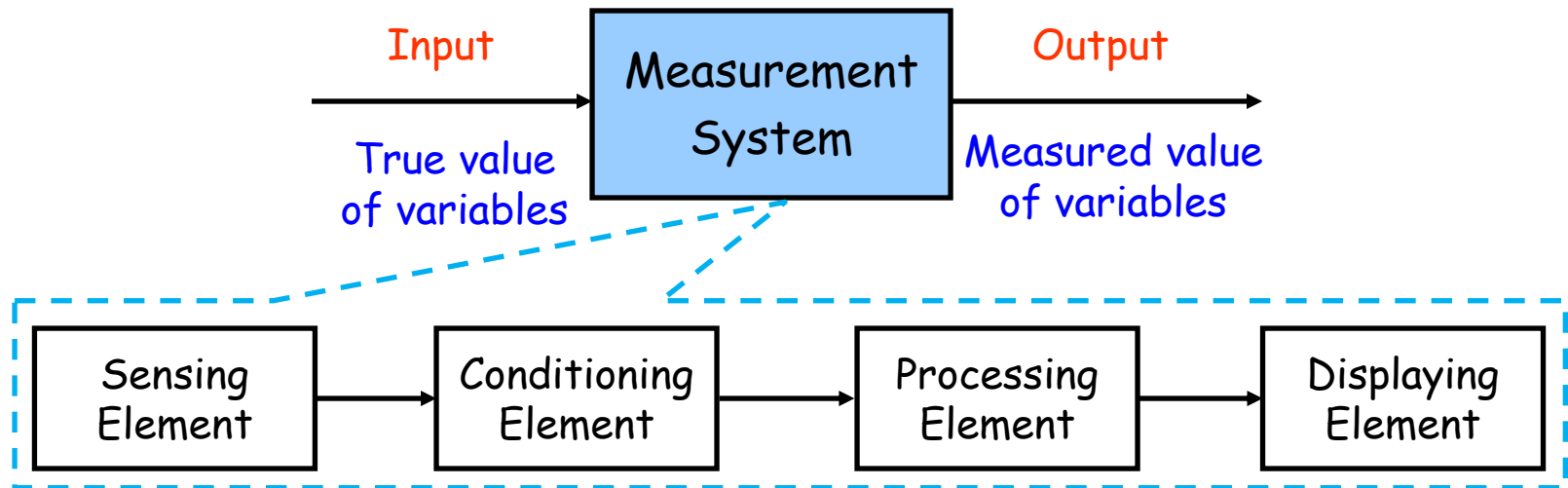
- We infer it from instrument output.
- Measurement = inference under uncertainty.

## > Formal definition:

- Measurement = Assigning a numerical value to a physical quantity using a defined procedure.
- E.g., length by a ruler (direct/indirect comparisons)

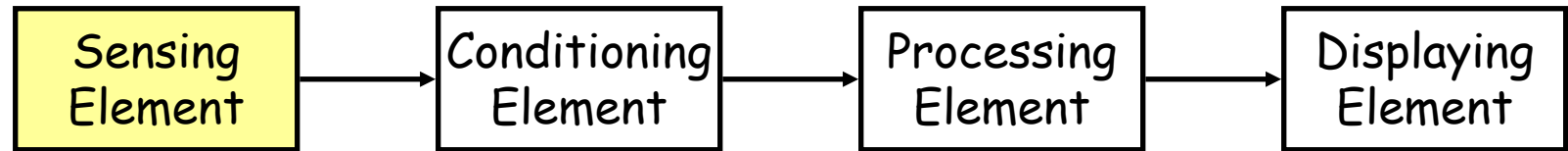
# Measurement System

- > Tool for quantifying the measured variable
  - How can a system be *devised* and *used* to provide reliable measurement data?
- > Essential elements:
  - Each element can distort reality



# Sensing Element

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## > Sensor:

- Converts physical phenomenon into measurable signal
- Usually in contact with the information carrier or medium
- E.g., mercury thermometer

## > Transducer:

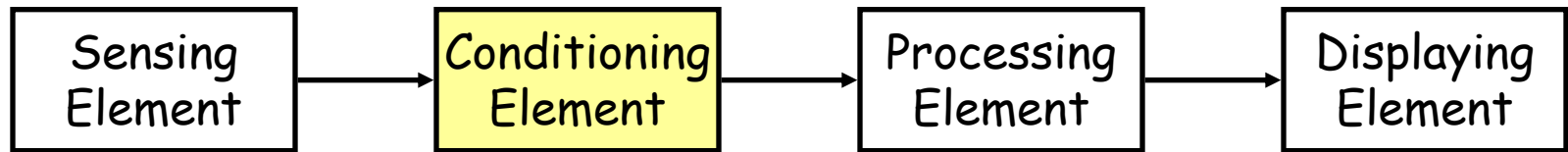
- Converts energy from one form to another
- Usually includes a sensor plus signal conditioning circuitry.
- E.g., thermocouple (senses temperature & converts thermal energy → voltage)

## > Sensor selection, placement, and installation are important

- E.g. 40°C “in the shade” ≠ 40°C “under the sun”

# Signal Conditioning

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## > Purpose:

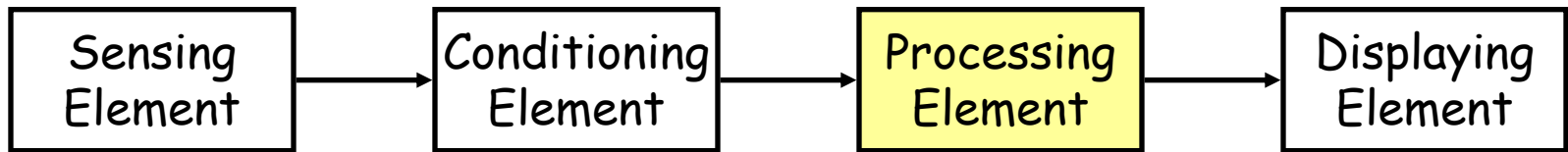
- Amplify
- Filter
- Convert format

## > Examples:

- Wheatstone bridge
- Operational amplifier
- ADC

# Processing

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## > Purpose:

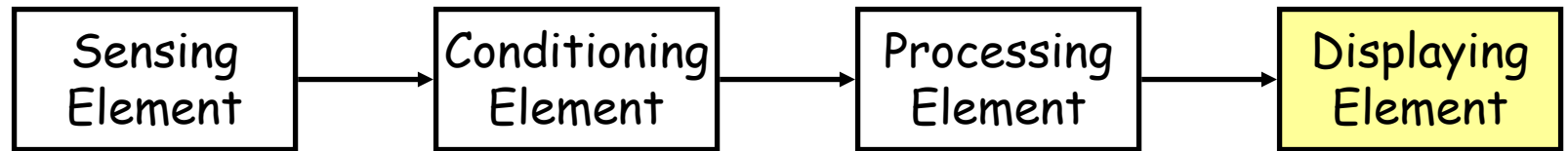
- Converting conditioned output into forms more suitable for presentation
- Calculating secondary variable from measurable variables

## > Examples:

- Digital filtering
- Compensation
- Calibration curve application

# Display

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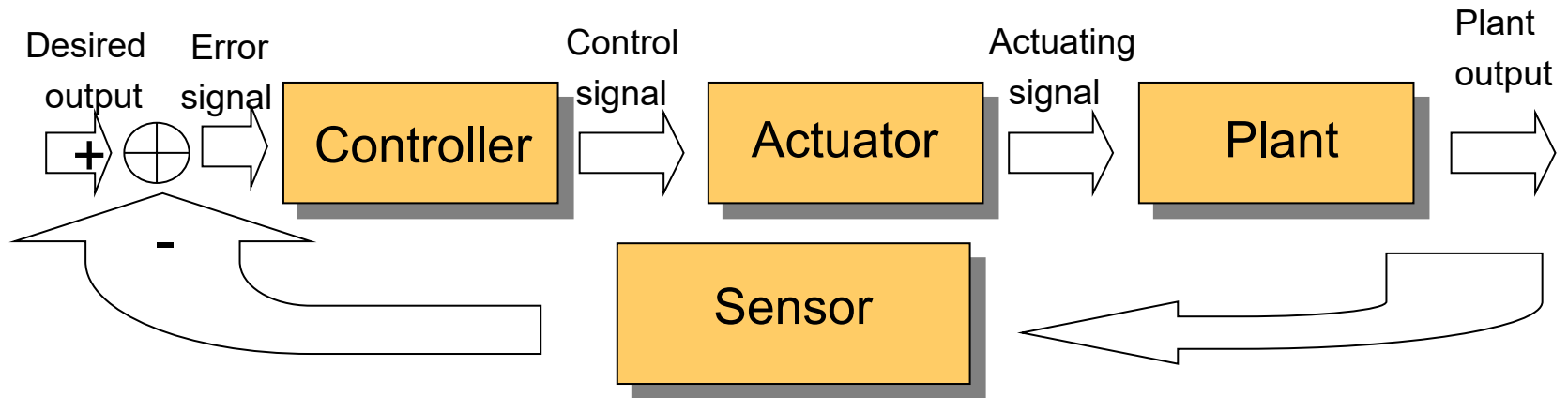
## > Purpose:

- Display and/or store measured signals in recognizable form (analog/digital)

## > Examples:

- Oscilloscope
- Digital display
- Data logger
- Cloud database

# Measurement in Control Systems



> Measurement enables feedback

- No measurement → no control.

# Measurement Error

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- > Ideal vs. Real measurement
  - Ideal: Measured = True
  - Real: Measured  $\neq$  True
  - Difference = Error
- > Formal definition: Error=Measured–True
  - True value usually unknown
  - Error does not imply that there is any mistake in the measurement process, although mistakes can cause errors
- > Two fundamental types of error
  - Systematic error (Bias)
  - Random error (Scatter)

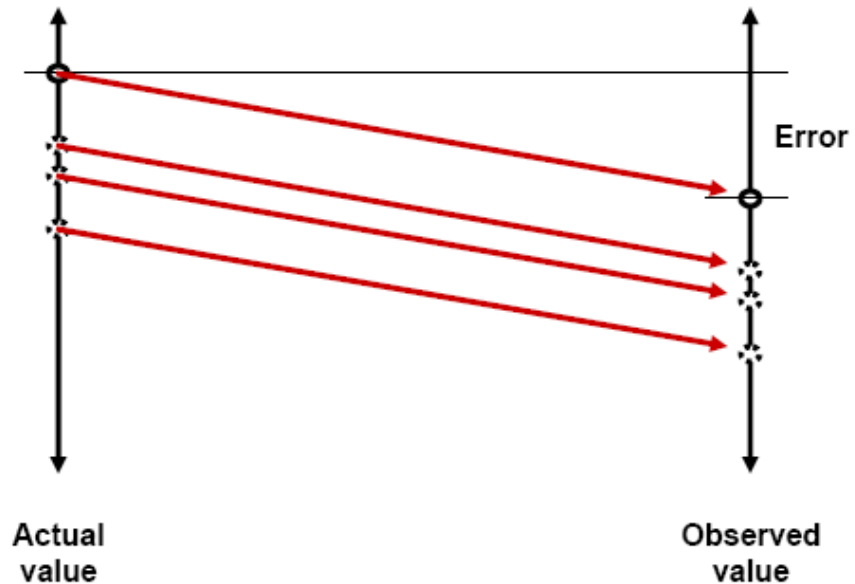
# Systematic Error

## > Characteristics:

- Repeatable
- Consistent offset
- Same direction

## > Sources:

- Calibration
- Nonlinearity
- Loading effect



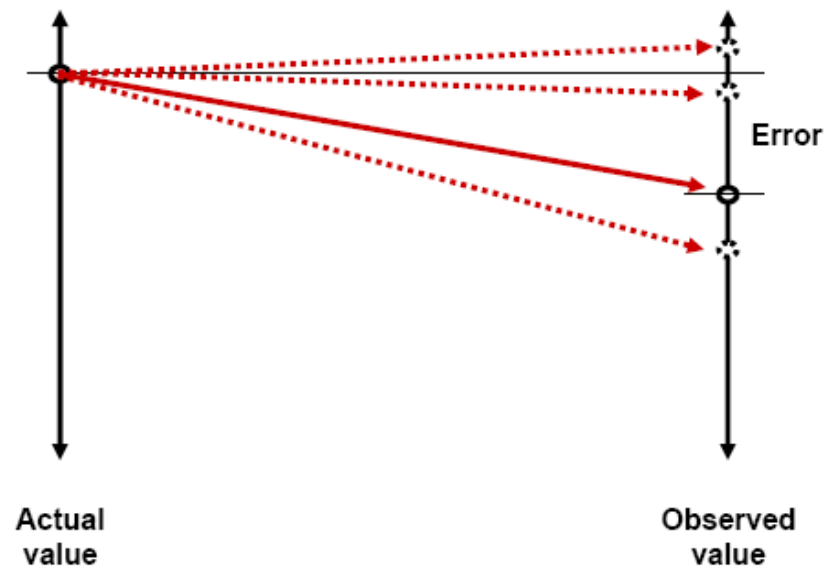
# Random Error

## > Characteristics:

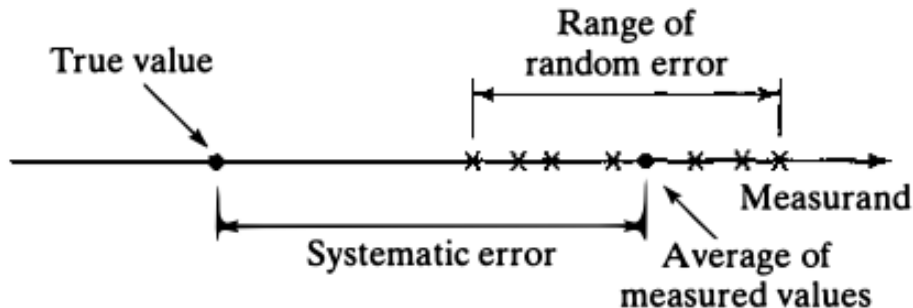
- Unpredictable
- Varies each measurement
- Statistical in nature

## > Sources:

- Noise
- Temperature variation
- Electrical interference



# Visualizing Error



- > Systematic error → shifts all data
  - Systematic error = average of readings - true value
  - Usually reduced by calibration or analytical correction of the data
  - Relates to **accuracy**
- > Random error → spreads data
  - Minimized by eliminating uncontrolled variables or statistical analysis
  - Relates to **precision**

# Accuracy vs. Precision

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- > Accuracy = closeness to true value
  - Often specified as a percentage of full-scale output
  - E.g., a device's range is from 0 to 5 V and the accuracy is  $\pm 5\%$  of full scale  $\rightarrow$  the uncertainty is  $\pm 0.25$  V, regardless of the reading.
- > Precision = repeatability
- > A system can be:
  - Precise but inaccurate
  - Accurate but not precise

## Example

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- > In a calibration test, 10 measurements have been made of the voltage of a battery that is known to have a true voltage of 6.11V. The readings are: 5.98, 6.05, 6.10, 6.06, 5.99, 5.96, 6.02, 6.09, 6.03, 5.99. Estimate the systematic & maximum random errors.
- > **Solution:**
  - The average of the 10 readings: 6.03V
  - systematic error = average value - true value =  $6.03 - 6.11 = -0.08\text{V}$
  - The reading that deviates the most from the average reading: 5.96V → The maximum random error =  $5.96 - 6.03 = -0.07\text{V}$ .
  - This simple method may not adequately describe random errors in a measuring system as it may be based on a single bad reading.

# Maximum Random Error vs. Statistical Error

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- > Max deviation may:
  - Depend on single bad reading
  - Not represent statistical uncertainty
- > The concept of confidence
  - Uncertainty: Estimated bounds of error with stated confidence
  - E.g. the uncertainty of a voltage measurement is  $\pm 1$  volt with 95% confidence
- > Think question:
  - If uncertainty is  $\pm 1V$ , does it mean measurement is wrong?
  - No. It defines confidence interval.

# Error Propagation

- > If the determination of  $y$  requires several measurements to be made & their values are put into the equation  $y = f(x_1, x_2, \dots, x_n)$ 
  - How do we determine the consequential error?

- > **Theorem:** 
$$\Delta y = \sqrt{\left(\frac{\partial y}{\partial x_1} \Delta x_1\right)^2 + \left(\frac{\partial y}{\partial x_2} \Delta x_2\right)^2 + \dots + \left(\frac{\partial y}{\partial x_n} \Delta x_n\right)^2}$$

- Errors accumulate via root-sum-square
- > **Example:** If the potential difference has been measured as  $2.1 \pm 0.2\text{V}$  and the current measured as  $0.25 \pm 0.01\text{A}$ , what will be the error in the resistance?
  - $R = \frac{U}{I} \rightarrow \Delta R = \pm \sqrt{\frac{U^2(\Delta I)^2 + I^2(\Delta U)^2}{I^4}} = \pm 0.87\Omega$
  - $R = 8.40 \pm 0.87\Omega$

# Range, Span, and Resolution of a System

## > Range:

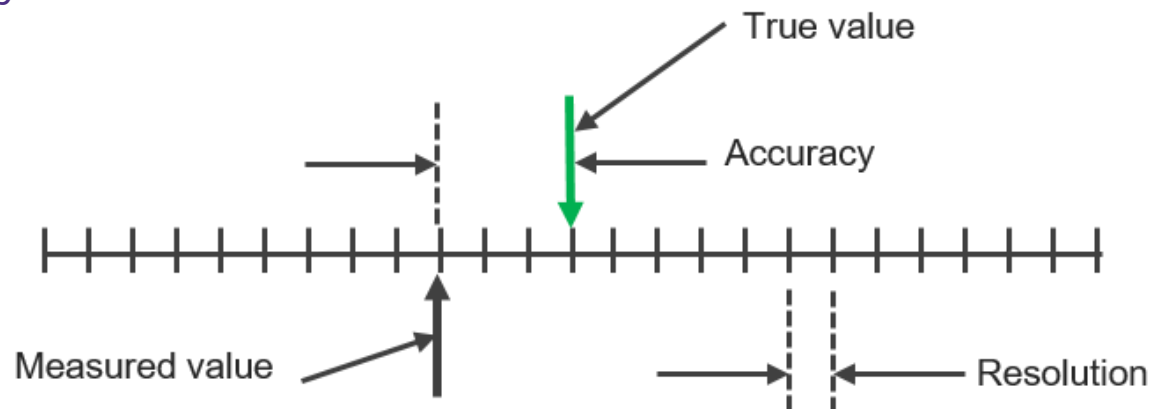
- Minimum to maximum measurable value

## > Span:

- Difference between the upper and lower values of the range

## > Resolution:

- Smallest detectable change:  $\text{Resolution (\%)} = \frac{\Delta x}{x_{\max} - x_{\min}} \times 100$
- Usually treated as a random error

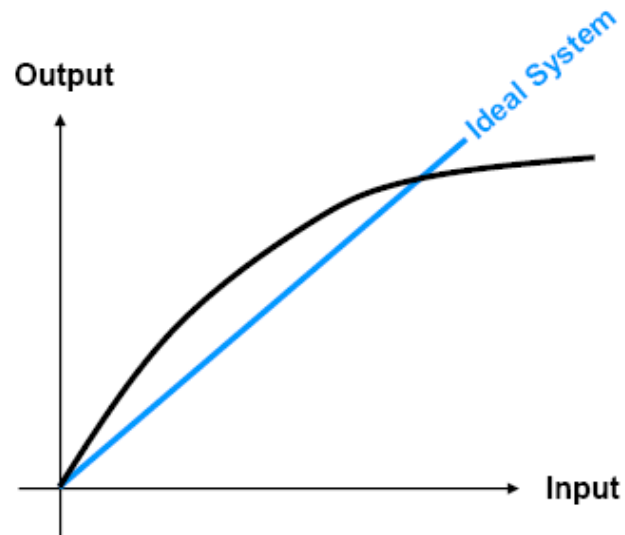


# Linearity

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## Measurement Error Characteristics

- Non-linearity



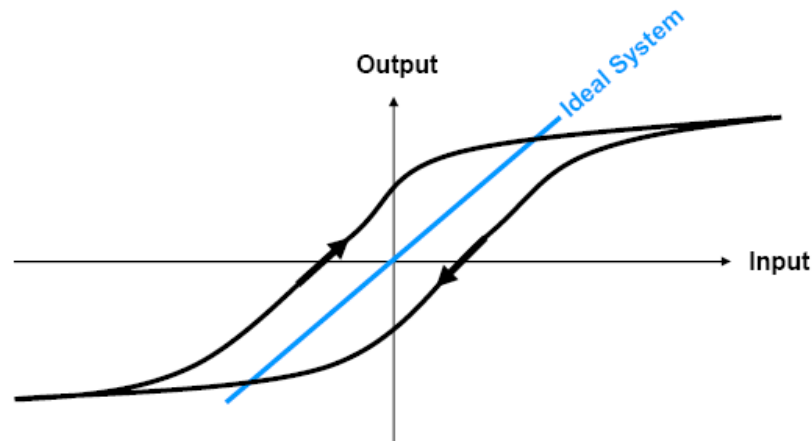
> Nonlinearity introduces systematic error

# Hysteresis

> Output depends on direction of input change

Measurement Error Characteristics

- Hysteresis



> Common in:

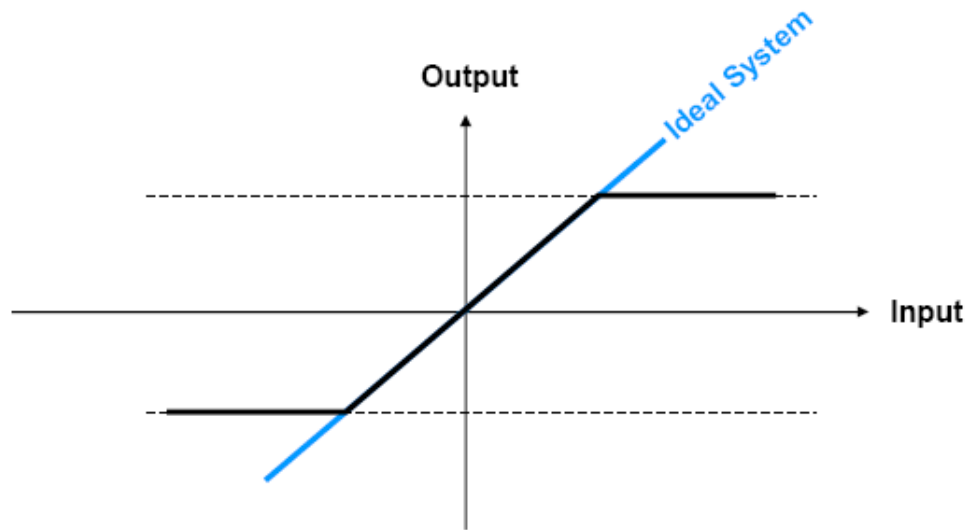
- Mechanical systems
- Magnetic materials

# Threshold & Dead Zone

> Threshold: Minimum input before output changes

## Measurement Error Characteristics

- Range-limitation (clipping)



> Dead zone: Range where output does not respond

# Drift and Sensitivity

> Drift: Output changes over time even if input constant.

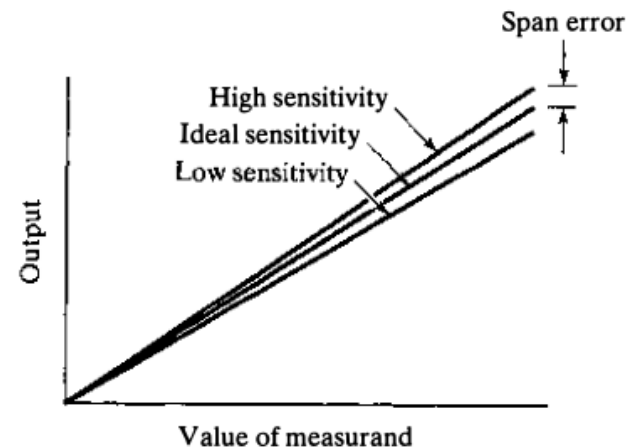
> Causes:

- Aging
- Temperature
- Component degradation

> Sensitivity:  $K = \frac{d(output)}{d(input)} \approx \frac{\Delta output}{\Delta input}$

> Higher sensitivity:

- Larger output change
- More detectable



# Example

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> A tachometer has

- Range 0–5000 rpm
- Accuracy  $\pm 5\%$  FS

When the shaft speed is zero, the device has a reading of 200rpm.  
What's the maximum error when reading a speed of 3500rpm?

> **Solution:**

- The accuracy specification indicates an uncertainty of  $\pm 0.05 \times 5000 = \pm 250$  rpm  $\rightarrow$  all readings have at least this uncertainty.
- There is a zero offset of 200 rpm in addition to the accuracy uncertainty
- Thus, the reading might be as much as  $250 + 200 = 450$  rpm high.
- The error estimate can be reduced if the zero is adjusted or if the data are corrected (by subtracting 200 rpm from the reading)

# Static Calibration Procedure

## > Steps

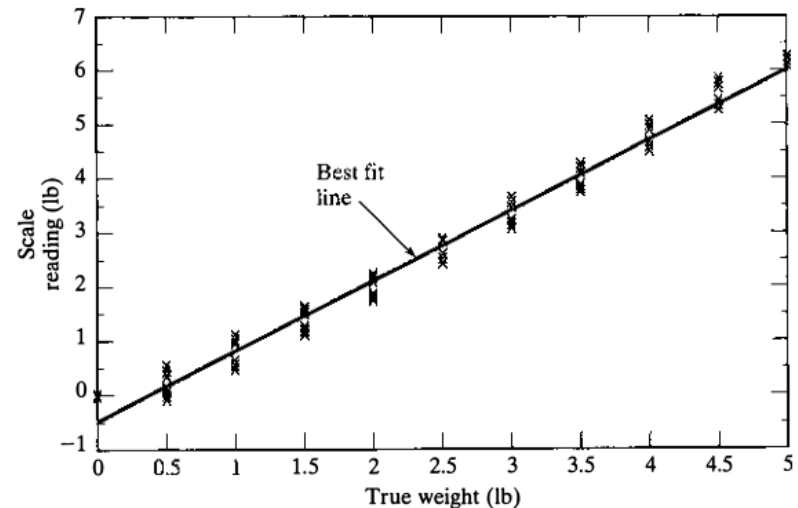
- Apply known input
- Record output
- Plot curve
- Fit model
- Compute residual error

## > Calibration curve

- Linear fit:  $y = ax + b$
- Correct measurement:  $x = \frac{y-b}{a}$



| True weight (lb) | Scale reading |         |         |         |         |         |
|------------------|---------------|---------|---------|---------|---------|---------|
|                  | Cycle 1       | Cycle 2 | Cycle 3 | Cycle 4 | Cycle 5 | Cycle 6 |
| 0.5              |               | 0.2     | 0.08    | 0.17    | 0.19    | 0.11    |
| 1                |               | 0.7     | 0.78    | 0.64    | 0.61    | 0.7     |
| 1.5              |               | 1.18    | 1.26    | 1.25    | 1.24    | 1.23    |
| 2                |               | 1.81    | 1.93    | 1.81    | 1.93    | 1.88    |
| 2.5              | 2.62          | 2.49    | 2.46    | 2.46    | 2.58    | 2.53    |
| 3                | 3.15          | 3.18    | 3.24    | 3.28    | 3.13    |         |
| 3.5              | 3.9           | 3.84    | 3.86    | 3.97    | 3.96    |         |
| 4                | 4.59          | 4.71    | 4.61    | 4.6     | 4.6     |         |
| 4.5              | 5.41          | 5.35    | 5.49    | 5.46    | 5.39    |         |
| 5                | 6.24          | 6.27    | 6.1     | 6.24    | 6.16    |         |
| 4.5              | 5.71          | 5.74    | 5.78    | 5.87    | 5.82    |         |
| 4                | 4.96          | 5.11    | 5.08    | 5.03    | 5.03    |         |
| 3.5              | 4.22          | 4.34    | 4.21    | 4.22    | 4.24    |         |
| 3                | 3.57          | 3.64    | 3.66    | 3.55    | 3.67    |         |
| 2.5              | 2.98          | 2.86    | 2.98    | 2.98    | 2.94    |         |
| 2                | 2.22          | 2.23    | 2.26    | 2.29    | 2.26    |         |
| 1.5              | 1.57          | 1.7     | 1.69    | 1.63    | 1.57    |         |
| 1                | 1.07          | 1.07    | 1.11    | 1.16    | 1.11    |         |
| 0.5              | 0.52          | 0.61    | 0.61    | 0.61    | 0.45    |         |
| 0                | 0.02          | 0.08    | 0.08    | -0.03   | 0.06    |         |



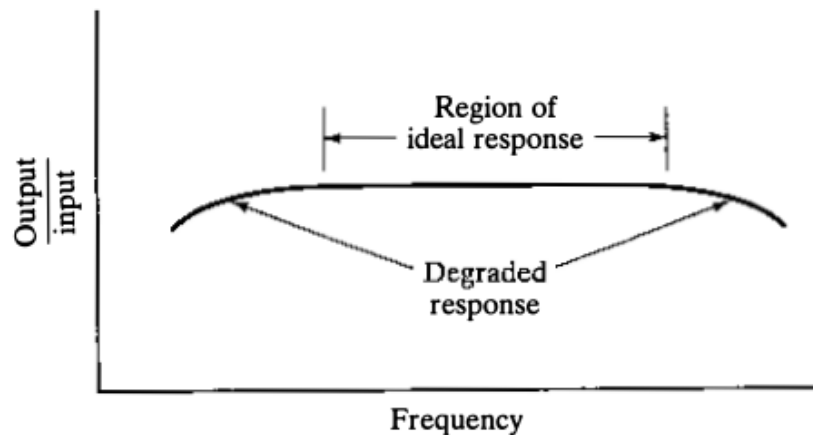
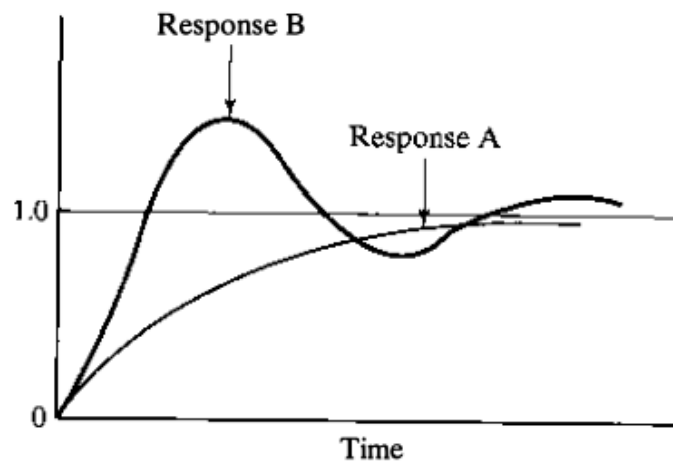
# Dynamic Calibration

## > Required when:

- Input changes with time
- Instrument has time constant

## > The experimenter must account for the *dynamic characteristics* of:

- the measuring system
- the the test system
- the interaction between the measuring system and the test system



# Reliability

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## > Reliability:

- Probability system meets accuracy under given conditions for given time.
- The reliability of a measurement system is likely to change with time

## > Failure rate: approximate model $R(t) = e^{-\lambda t}$

- $\lambda$ : failure rate
- $R(t)$ : The probability that the system is still functioning correctly at time  $t$
- A high reliability system will have a low failure rate.

## > **Quiz:** The failure rate for a pressure measurement system is 1% per year. What is the average number of reliable measurements if the system is used 300 times a year?

- *Ans:* 297

# Error Reduction Strategies

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## > Systematic:

- Calibration
- Compensation
- Proper design

## > Random:

- Shielding
- Filtering
- Averaging

## > **Design Insight**

- Perfect measurement is impossible
- Engineering goal: Minimize uncertainty to acceptable level

# Summary

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## Key takeaways

- > Measurement is inference.
- > Error is unavoidable.
- > Bias and noise behave differently.
- > Calibration corrects bias.
- > Statistics handles random uncertainty.

# NEXT LECTURE

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## [Flipped class] Functions and pointers

### > Pre-class

- K&R “The C Programming Language”, Chapters 1 & 2 (p. 9-52)
- Michael Kerrisk, “The Linux Programming Interface: A Linux and UNIX System Programming Handbook”, Chapter 7 (p. 139-152)

### > In class

- Reinforcement/enrichment discussion
- Quizzes

### > Post class

- Homework
- Consultation (if needed)